

Rank-one irreducible perturbations from cyclic approximate eigenvectors

Partial-result packet for arXiv:2504.17190

candidate partial result—likely valid

1. Source problem and the unresolved bottleneck

For a separable infinite-dimensional complex Hilbert space $\mathcal{H}$, Fang–Jiang–Ma–Shen–Shi–Wang ask:

For every $T \in \mathcal{B}(\mathcal{H})$ and $\varepsilon > 0$, is there a trace-class K with $\|K\|_1 < \varepsilon$ such that $T + K$ is irreducible?

This is **Problem A** in arXiv:2504.17190v6. Their Main Theorem reduces it to their type-II₁ factor conjecture: a generator of a type-II₁ factor should admit an arbitrarily small trace-class perturbation that is a direct sum of at most countably many irreducible operators. The source proves many large subcases but leaves this endpoint open.

The result below gives a different, rank-one sufficient condition. It does not solve the type-II₁ conjecture, because cyclicity for the von Neumann algebra $W^*(T)$ is much weaker than one-sided cyclicity for T or T^* .

We use $u \otimes v$ for the rank-one operator $x \mapsto \langle x, v \rangle u$. Let $\Omega_\infty(T)$ be the unbounded component of the resolvent set $\rho(T)$.

2. Main partial theorem

Theorem 1 (cyclic approximate-eigenvector criterion). *Let $T \in \mathcal{B}(\mathcal{H})$. Suppose that some $\lambda \in \partial\Omega_\infty(T) \cap \sigma(T)$ admits unit vectors ξ_j such that*

1. *every ξ_j is cyclic for T^* , and*
2. $\|(T - \lambda)\xi_j\| \rightarrow 0$.

Then for every $\varepsilon > 0$ there is a rank-one operator K with $\|K\|_1 < \varepsilon$ such that $T + K$ is irreducible.

Corollary 2. *If the cyclic vectors for T^* are dense in $\mathcal{H}$, then T satisfies the conclusion of Theorem 1. The same conclusion holds if the cyclic vectors for T are dense.*

The second assertion follows by applying the first assertion to T^* and taking adjoints. Thus this provides a full trace-class-density answer for every operator in either of these co-cyclic classes, and the perturbation is in fact rank one.

3. Proof

Fix $\varepsilon > 0$. Choose a unit cyclic vector ξ for T^* with $\|(T - \lambda)\xi\|$ as small as needed below. On $\Omega_\infty(T)$ consider

$$g_\xi(z) = \langle (T - z)^{-1}\xi, \xi \rangle.$$

This analytic function is not identically zero, since $g_\xi(z) = -z^{-1} + O(|z|^{-2})$ as $z \rightarrow \infty$. Its zeros are therefore isolated. We may choose $\mu \in \Omega_\infty(T)$ arbitrarily close to λ such that

$$g_\xi(\mu) \neq 0, \quad \|(T - \mu)\xi\| < \varepsilon.$$

Put $P = \xi \otimes \xi$ and

$$K = (\mu I - T)P, \quad S = T + K.$$

Then K has rank one and $\|K\|_1 = \|(T - \mu)\xi\| < \varepsilon$. Moreover,

$$S - \mu = (T - \mu)(I - P). \tag{1}$$

Consequently $\ker(S - \mu) = \mathbb{C}\xi$ and $\text{Ran}(S - \mu) = (T - \mu)\xi^\perp$. The choice $g_\xi(\mu) \neq 0$ says exactly that

$$(T - \mu)^{-1}\xi \notin \xi^\perp,$$

or equivalently $\xi \notin \text{Ran}(S - \mu)$. Thus the eigenvalue μ is semisimple with one-dimensional eigenspace.

It is also isolated. Indeed, on $\Omega_\infty(T)$ the possible new spectrum of the rank-one perturbation is the zero set of the analytic Fredholm determinant

$$d(z) = \det(I + (T - z)^{-1}K).$$

This determinant tends to 1 at infinity and hence is not identically zero; its zeros are isolated. Equation (1) shows $d(\mu) = 0$. Therefore the Riesz projection E of S at μ has rank one and range $\mathbb{C}\xi$.

The Riesz projection belongs to $W^*(S)$. Hence the orthogonal range projection

$$P = s(EE^*)$$

also belongs to $W^*(S)$. On the other hand, K^* has range contained in $\mathbb{C}\xi$. Induction gives, for every $n \geq 0$,

$$\text{span}\{\xi, S^*\xi, \dots, (S^*)^n\xi\} = \text{span}\{\xi, T^*\xi, \dots, (T^*)^n\xi\}.$$

Thus ξ is cyclic for S^* . Finally, for all $m, n \geq 0$,

$$(S^*)^m P S^n = (S^*)^m \xi \otimes (S^*)^n \xi \in W^*(S).$$

Because the S^* -orbit of ξ is dense, these rank-one operators generate $\mathcal{B}(\mathcal{H})$ as a von Neumann algebra. Hence $W^*(S) = \mathcal{B}(\mathcal{H})$, so $S = T + K$ is irreducible. This proves Theorem 1.

For Corollary 2, every point of $\partial\Omega_\infty(T) \cap \sigma(T)$ is an approximate eigenvalue of T . Approximate its unit approximate eigenvectors by cyclic vectors for T^* ; after normalization the residuals still tend to zero. The theorem applies.

4. Relation to the type-II₁ conjecture

The proof isolates a clean mechanism:

$$\begin{aligned} & \text{cyclic approximate eigenvector} \\ & \implies \text{small rank-one perturbation with a recoverable simple eigenprojection} \\ & \implies W^*(T + K) = \mathcal{B}(\mathcal{H}). \end{aligned}$$

The first implication is quantitative: the trace norm of the perturbation is exactly $\|(T - \mu)\xi\|$.

For a cyclic representation of a type-II₁ factor, a vector cyclic for $W^*(T)$ need only be cyclic under words in T and T^* ; it need not be cyclic under the powers of T^* alone. This is the precise point at which the rank-one proof stops short of the source conjecture.

5. Focused upgrade attempts and obstruction audit

Six upgrade routes were tested after obtaining Theorem 1.

1. **Cyclic factor representation $\Rightarrow$ one-sided cyclicity.** This implication fails already in the finite-factor model: on $L^2(M_n)$, left multiplication by a $*$ -generator of M_n generates a cyclic factor representation, but every eigenvalue has the right-module multiplicity and the operator need not be cyclic.
2. **Finite cyclic multiplicity.** Replacing one simple eigenvalue by a finite cluster can recover a finite-dimensional Riesz corner, but a trace-norm estimate requires a cyclic tuple consisting of simultaneous small-residual vectors. Von Neumann-algebra cyclicity does not supply such a tuple.
3. **Countably many rank-one eigenvalue insertions.** A countable cyclic set always exists, but nuclearity requires the sum of all residual norms to be finite and arbitrarily small. There is no mechanism forcing a total cyclic set into a shrinking approximate-eigenspace.
4. **First make T^* cyclic, then apply the theorem.** Norm-density results for cyclic operators do not provide perturbations small in trace norm. The same endpoint ideal obstruction reappears, so this route would merely replace Problem A by another trace-class density problem.
5. **Recover a rank-one projection without inserting an eigenvalue.** Adding $i\delta(\xi \otimes \xi)$ kills the old factor commutant when ξ is factor-cyclic, but the projection is not in general recoverable from the single perturbed operator. The eigenvalue construction solves recovery only at the cost of demanding one-sided cyclicity.
6. **Relative-normalizer completion.** The source’s averaging lemma succeeds when a diffuse subalgebra of $W^*(\text{Re } T)$ has enough relative normalizers. Singular masas and strongly solid factors show why such normalizers cannot be assumed in the remaining type-II₁ case.

The remaining gap is therefore structural, not an unverified estimate in the rank-one proof. A full upgrade would require either a new way to recover finite-rank data from $T + K$ without one-sided cyclicity, or a trace-class theorem producing a cyclic approximate-eigenvector configuration for arbitrary type-II₁ generators.

6. Status and novelty boundary

The theorem is a self-contained candidate partial result. Targeted searches of the cheap run indexes and arXiv for the combinations “rank-one perturbation”, “irreducible operator”, and “cyclic adjoint” found no exact match, but this is not a comprehensive literature proof of novelty. Human review should check classical rank-one perturbation and cyclic-operator literature before making a novelty claim.

The source’s full Problem A and its equivalent type-II₁ conjecture remain unresolved by this packet.